

DBMS – SELLER AND INVENTORY MANAGEMENT
TITAN EYE – EYEWEAR E-COMMERCE DATABASE SYSTEM

1. Database Design

1.1 Seller Table

Attribute	Data Type	Constraint
Seller_ID	NUMBER	Primary Key
Seller_Name	VARCHAR2(150)	NOT NULL
Email	VARCHAR2(150)	—
Phone_Number	VARCHAR2(15)	—
Address	VARCHAR2(250)	—

1.2 Inventory Table

Make another table:

Attribute	Data Type	Constraint
Inventory_ID	NUMBER	Primary Key
Seller_ID	NUMBER	Foreign Key
Product_ID	NUMBER	Foreign Key
Stock_Quantity	NUMBER	—
Product_Status	VARCHAR2(20)	—

2. Table Relationships

2. Table Relationships

The Seller and Inventory tables are related to maintain seller-wise product stock information. The Inventory table also connects with the existing Product table.



Relationships:

1 Seller → Inventory (One-to-Many)

- One seller can have multiple inventory records.
- Seller_ID in Inventory is a foreign key referencing Seller_ID in Seller.

2 Product → Inventory (One-to-Many)

- One product can have inventory information associated with it.
- Product_ID in Inventory is a foreign key referencing Product_ID in Product.

Foreign Key Constraints (from SQL):

```
ALTER TABLE Inventory
ADD CONSTRAINT fk_inventory_seller
FOREIGN KEY (Seller_ID)
REFERENCES Seller(Seller_ID);
```

```
ALTER TABLE Inventory
ADD CONSTRAINT fk_inventory_product
FOREIGN KEY (Product_ID)
REFERENCES Product(Product_ID);
```

3. Table Creation

3.1 Create Seller Table

```
SQL> CREATE TABLE Seller (
2     Seller_ID NUMBER PRIMARY KEY,
3     Seller_Name VARCHAR2(150) NOT NULL,
4     Email VARCHAR2(150),
5     Phone_Number VARCHAR2(15),
6     Address VARCHAR2(250)
7 );
```

Table created.

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (
2     Inventory_ID NUMBER PRIMARY KEY,
3     Seller_ID NUMBER,
4     Product_ID NUMBER,
5     Stock_Quantity NUMBER,
6     Product_Status VARCHAR2(20)
7 );
```

Table created.

3.3 Establish Foreign Key Relationships

```
SQL> DESC Inventory;
Name                                         Null?    Type
-----
INVENTORY_ID                                NOT NULL NUMBER
SELLER_ID                                   NUMBER
PRODUCT_ID                                 NUMBER
STOCK_QUANTITY                             NUMBER
PRODUCT_STATUS                             VARCHAR2(20)

SQL>
SQL> ALTER TABLE Inventory
  2  ADD CONSTRAINT fk_inventory_seller
  3  FOREIGN KEY (Seller_ID)
  4  REFERENCES Seller(Seller_ID);

Table altered.
```

```
SQL> ALTER TABLE Inventory
  2  ADD CONSTRAINT fk_inventory_product
  3  FOREIGN KEY (Product_ID)
  4  REFERENCES Product(Product_ID);

Table altered.

SQL> DESC Inventory;
Name                                         Null?    Type
-----
INVENTORY_ID                                NOT NULL NUMBER
SELLER_ID                                   NUMBER
PRODUCT_ID                                 NUMBER
STOCK_QUANTITY                             NUMBER
PRODUCT_STATUS                             VARCHAR2(20)
```

4. Seller Data Insertion

```
SQL> SELECT * FROM Seller;

no rows selected

SQL> INSERT INTO Seller
  2  VALUES (1, 'Titan Eye Seller', 'seller1@titaneyeye.com',
  3          '9876543210', 'Chennai');

1 row created.

SQL> INSERT INTO Seller
  2  VALUES (2, 'Vision Care Seller', 'seller2@visioncare.com',
  3          '9876501234', 'Bangalore');

1 row created.

SQL> COMMIT;

Commit complete.
```

```
SQL> SELECT Product_ID, Product_Name  
2 FROM Product;
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
-----  
101  
Titan Classic Frame
```

```
102  
Titan Metal Frame
```

```
103  
Titan Aviator Sunglasses
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
-----  
104  
Titan Round Sunglasses
```

5.Inventory Data Insertion

```
SQL> INSERT INTO Inventory  
2 VALUES (1002, 1, 102, 15, 'AVAILABLE');
```

```
1 row created.
```

```
SQL> INSERT INTO Inventory  
2 VALUES (1003, 2, 103, 10, 'AVAILABLE');
```

```
1 row created.
```

```
SQL>
```

```
SQL> COMMIT;
```

```
Commit complete.
```

```
SQL> SELECT * FROM Inventory;
```

INVENTORY_ID	SELLER_ID	PRODUCT_ID	STOCK_QUANTITY	PRODUCT_STATUS
1001	1	101	25	AVAILABLE
1002	1	102	15	AVAILABLE
1003	2	103	10	AVAILABLE

6. Inventory Update

6.1 Update Stock Quantity

```
SQL> UPDATE Inventory
  2  SET Stock_Quantity = 30
  3  WHERE Inventory_ID = 1001;
```

1 row updated.

```
SQL> COMMIT;
```

Commit complete.

```
SQL> SELECT * FROM Inventory;
```

INVENTORY_ID	SELLER_ID	PRODUCT_ID	STOCK_QUANTITY	PRODUCT_STATUS
1001	1	101	30	AVAILABLE
1002	1	102	15	AVAILABLE
1003	2	103	10	AVAILABLE

6.2 Update Product Status

```
SQL> UPDATE Inventory
  2  SET Stock_Quantity = 0,
  3      Product_Status = 'UNAVAILABLE'
  4  WHERE Product_ID = 103;
```

1 row updated.

```
SQL>
```

```
SQL> COMMIT;
```

Commit complete.

```
SQL> SELECT *
  2  FROM Inventory
  3  WHERE Product_Status = 'AVAILABLE';
```

INVENTORY_ID	SELLER_ID	PRODUCT_ID	STOCK_QUANTITY	PRODUCT_STATUS
1001	1	101	30	AVAILABLE
1002	1	102	15	AVAILABLE

```
SQL> SELECT *
  2  FROM Inventory
  3  WHERE Product_Status = 'UNAVAILABLE';
```

INVENTORY_ID	SELLER_ID	PRODUCT_ID	STOCK_QUANTITY	PRODUCT_STATUS
1003	2	103	0	UNAVAILABLE


```

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
      1
Titan Eye Seller
      102

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
Titan Metal Frame
      15 AVAILABLE

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
      2

```

```

Vision Care Seller
      103

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
Titan Aviator Sunglasses
      0 UNAVAILABLE

```

8. Inventory Status

8.1 Display Available Products

```

SQL> SELECT
  2     s.Seller_Name,
  3     p.Product_Name,
  4     i.Stock_Quantity,
  5     i.Product_Status
  6 FROM Seller s
  7 JOIN Inventory i
  8     ON s.Seller_ID = i.Seller_ID
  9 JOIN Product p
 10     ON i.Product_ID = p.Product_ID
 11 WHERE i.Product_Status = 'AVAILABLE';

```

SELLER_NAME

PRODUCT_NAME

STOCK_QUANTITY PRODUCT_STATUS

Titan Eye Seller

Titan Classic Frame

30 AVAILABLE

Titan Eye Seller

Titan Metal Frame

15 AVAILABLE

SELLER_NAME

PRODUCT_NAME

STOCK_QUANTITY PRODUCT_STATUS

8.2 Display Unavailable Products

```

SQL> SELECT
  2     s.Seller_Name,
  3     p.Product_Name,
  4     i.Stock_Quantity,
  5     i.Product_Status
  6 FROM Seller s
  7 JOIN Inventory i
  8     ON s.Seller_ID = i.Seller_ID
  9 JOIN Product p
 10     ON i.Product_ID = p.Product_ID
 11 WHERE i.Product_Status = 'UNAVAILABLE';

```

SELLER_NAME

PRODUCT_NAME

STOCK_QUANTITY PRODUCT_STATUS

Vision Care Seller

Titan Aviator Sunglasses

0 UNAVAILABLE

```
SQL> SELECT
  2     s.Seller_Name,
  3     SUM(i.Stock_Quantity) AS Total_Stock
  4 FROM Seller s
  5 JOIN Inventory i
  6     ON s.Seller_ID = i.Seller_ID
  7 GROUP BY s.Seller_Name;
```

SELLER_NAME

TOTAL_STOCK

Titan Eye Seller
45

Vision Care Seller
0